Name:- Yash Sandip Mali

Roll No:- C-08

Lab Assignment:- (Group-A) 1.a)

Code:-

```
#include <stdio.h>
int linearSearch(int arr[], int n, int key){
    int i;
    for(i = 0; i < n; i++){
        if(arr[i] == key {
            return i; // Roll number found
        }
    }
    return -1;// Roll number not found
}

int main(){
    int roll[100];
    int n, i, choice, key, result;
    printf("Enter the number of students attended training: ");
    scanf("%d", &n);
    printf("Enter roll numbers:\n");
    for(i = 0; i < n; i++){
        scanf("%d", &roll[i]);
    }
    do{
        printf("\n===== MENU =====\n");
        printf("1. Display Roll Numbers\n");
        printf("2. Search Roll Number\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
```

```
        switch(choice){
            case 1:
                printf("\nStudents Attended Training:\n");
                for(i = 0; i < n; i++){
                    printf("%d ", roll[i]);
                }
                printf("\n");
                break;

            case 2:
                printf("Enter roll number to search: ");
                scanf("%d", &key);
                result = linearSearch(roll, n, key);
                if(result != -1){
                  printf("Student with Roll Number %d attended the
training program.\n", key);
                }
                else{
                   printf("Student with Roll Number %d did NOT attend
the training program.\n", key);
                }
                break;
            case 3:
                printf("Exiting program...\n");
                break;
            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while(choice != 3);

    return 0;
}
```

Output:-

```
Enter the number of students attended training: 5
Enter roll numbers:
65
76
90
43
21
===== MENU =====
1. Display Roll Numbers
2. Search Roll Number
3. Exit
Enter your choice: 1

Students Attended Training:
65 76 90 43 21

===== MENU =====
1. Display Roll Numbers
2. Search Roll Number
3. Exit
Enter your choice: 2
Enter roll number to search: 21
Student with Roll Number 21 attended the training program.

===== MENU =====
1. Display Roll Numbers
2. Search Roll Number
3. Exit
Enter your choice: 3
Exiting program...
```

Name:- Yash Sandip Mali

Roll No:- C-08

Lab Assignment:- (Group-A) 1.b)

Code:-

```
#include <stdio.h>
int binarySearch(int arr[], int n, int key){
    int low = 0, high = n - 1, mid;
    while(low <= high){
        mid = (low + high) / 2;
        if(arr[mid] == key){
            return mid;   // Roll number found
        }
        else if(arr[mid] < key){
            low = mid + 1;
        }
        else{
            high = mid - 1;
        }
    }
    return -1;   // Roll number not found
}
int main(){
    int roll[100];
    int n, i, choice, key, result;
    printf("Enter the number of students attended training: ");
    scanf("%d", &n);
    printf("Enter roll numbers:\n");
    for(i = 0; i < n; i++){
        scanf("%d", &roll[i]);
    }
    do{
        printf("\n===== MENU =====\n");
```

```
        printf("1. Display Roll Numbers\n");
        printf("2. Search Roll Number\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice){
            case 1:
                printf("\nStudents Attended Training:\n");
                for(i = 0; i < n; i++){
                    printf("%d ", roll[i]);}
                printf("\n");
                break;
            case 2:
                printf("Enter roll number to search: ");
                scanf("%d", &key);
                result = binarySearch(roll, n, key);
                if(result != -1){
                  printf("Student with Roll Number %d attended the
training program.\n", key);
                }
                else{
                   printf("Student with Roll Number %d did NOT attend
the training program.\n", key);}
                break;
            case 3:
                printf("Exiting program...\n");
                break;
            default:
                printf("Invalid choice! Please try again.\n");}
    } while(choice != 3);
 return 0;
    }
```

Output:-

```
Enter the number of students attended training: 5
Enter roll numbers:
56
34
78
23
98
===== MENU =====
1. Display Roll Numbers
2. Search Roll Number
3. Exit
Enter your choice: 1

Students Attended Training:
56 34 78 23 98

===== MENU =====
1. Display Roll Numbers
2. Search Roll Number
3. Exit
Enter your choice: 2
Enter roll number to search: 45
Student with Roll Number 45 did NOT attend the training program.

===== MENU =====
1. Display Roll Numbers
2. Search Roll Number
3. Exit
Enter your choice: 3
Exiting program...
```